

Predictive Analytics of Adolescent Behavioral Patterns for Mental Health Forecasting

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Abstract—Mental health issues have become a pressing global concern, affecting a growing number of adolescents. Many individuals experience psychological distress without timely recognition, as traditional assessment methods relying heavily on subjective self-reports that often suffer from bias, delayed detection, and underreporting. This gap leads to exacerbated mental health deterioration, impacting quality of life, academic performance, and social functioning, underscoring the urgent need for objective, real-time prediction tools. The study addresses this need by developing a multimodal machine learning framework for early mental state forecasting, with significant implications for enabling proactive interventions and alleviating healthcare burdens. The study utilizes a dataset that integrate diverse mental health factors including demographics, stress triggers, performance metrics (CGPA), satisfaction scores, and lifestyle data. For methodology, the study first preprocesses data through normalization, missing value imputation, and feature extraction to identify behavioral patterns. Then, the study compares four types of machine learning models: logistic regression, support vector machine (SVM), random forest, and XGBoost. Evaluates the model performances based on accuracy, precision, recall, F1-score, and AUC-ROC. Preliminary results show the proposed framework outperforms baseline models, with XGBoost achieving the highest AUC-ROC of 0.9610, demonstrating strong capability in distinguishing mental health states. These findings validate the potential of multimodal data and optimized machine learning algorithms in mental health prediction. Looking forward, the study aims to expand datasets with more diverse populations, refine feature engineering, and enhance real-world applicability, striving to develop a scalable system that improves adolescent mental health monitoring and intervention efficacy.

Keywords—Behavioral pattern mining, Deep learning, Predictive modeling, Mental health, Proactive intervention

I. INTRODUCTION

Global mental health issues have reached epidemic proportions, with approximately one in nine individuals experiencing psychological distress—a phenomenon particularly prevalent among adolescents. This underscores the urgency for effective early detection and intervention strategies. The World Health Organization emphasizes that untreated mental health problems severely impact quality of life and impose heavy economic burdens on healthcare systems. However, traditional assessment methods relying on self-reports suffer from subjectivity, bias, and poor early detection, as many remain unaware of issues or avoid seeking help.

Predictive models for mental states offer a solution by analyzing multi-source behavioral data from daily activities (e.g., academic performance, sleep patterns) to identify subtle patterns of emerging mental health issues, enabling objective,

real-time assessments and proactive intervention. Multimodal data integrating behavioral, academic, and lifestyle information provides a more comprehensive view than single-source data.

Recent progress includes Jamin Patel et al.'s review of predictive AI in mHealth for youth mental health, noting multimodal data (wearables/smartphones) and key predictors (usage, sleep, activity) aid forecasting, with challenges like small samples and privacy issues, and models lacking human-centered design [1]. T. Bellido Bel et al.'s pilot uses app data, combining active/passive inputs to predict adolescent mental health risk, with improved accuracy (0.67 balanced, AUC 0.7) and step count/questionnaire items as key features [2]. Muhammad Nouman et al. propose a text-based prediction framework, achieving 92.4% accuracy, outperforming other models [3]. Yet limitations persist, including insufficient data diversity, unresolved generalizability issues, and the demand for more accurate models—factors that underscore the importance of developing more robust frameworks [4-6].

This article focuses on addressing these limitations by designing a model that integrates academic metrics, lifestyle data (sleep duration, study hours), and psychological indicators, with preprocessing to extract key features such as behavioral patterns and emotional tendencies. It develops an optimized multimodal machine learning framework, comparing performance across traditional and ensemble models to identify the most effective predictor. The goal is to form a practical and scalable early prediction system to enable timely interventions and improve adolescent mental well-being.

To achieve this, the study utilizes a comprehensive dataset encompassing demographics (age, gender), stress triggers (academic pressure), performance indicators (CGPA, task completion rates), satisfaction scores (life/study contentment), and lifestyle metrics (sleep duration, daily screen time, physical activity). First, the study visualizes key relationships—such as correlations between sleep patterns and stress levels or CGPA trends across age groups—via heatmaps and scatter plots to identify critical features. After preprocessing (normalization, handling missing values), the study constructs and compares four machine learning models: logistic regression, SVM, random forest, and XGBoost. The study evaluates the model performances based on accuracy, precision, recall, F1-score, and AUC-ROC. The goal is to apply the model that demonstrates superior accuracy in cross-scenarios and captures multi-modal patterns for real-time mental state prediction.

II. METHODOLOGY

To achieve accurate mental state forecasting, the study involves several key steps including data integration, feature engineering, model construction, and result analysis.

A. Multi-aspect Data Integration

The first step is to obtain a clean dataset, as high-quality data forms the foundation for subsequent analysis and modeling. The initial data is sourced from Kaggle, but it contains both relevant and irrelevant features. Thus, it cannot be directly used for research due to potential issues such as irrelevant features, missing values, and inconsistent formatting.

To address this, the study starts by conducting a thorough review of the dataset. First, the study examines all variables in the initial data and carefully screen them against the research objectives, removing variables that are unrelated to the study. For example, eliminating redundant identifiers or attributes that do not contribute to the analysis of target metrics. Next, the study focuses on handling missing values, which are a common issue in raw datasets. The study checks the proportion and distribution of missing values for each remaining variable; for variables with a small number of missing values, the study uses appropriate methods such as mean/median imputation or mode filling to retain valuable information, while for those with excessive missing values that could bias the results, the study considers excluding these values.

After these steps of screening and processing, a clean dataset is constructed. These variables are selected to align with the research focus, ensuring that the dataset is both concise and informative.

The final dataset includes the following key fields:

Demographics & Academic Context: Gender, age, city, profession, academic pressure, CGPA, study satisfaction.

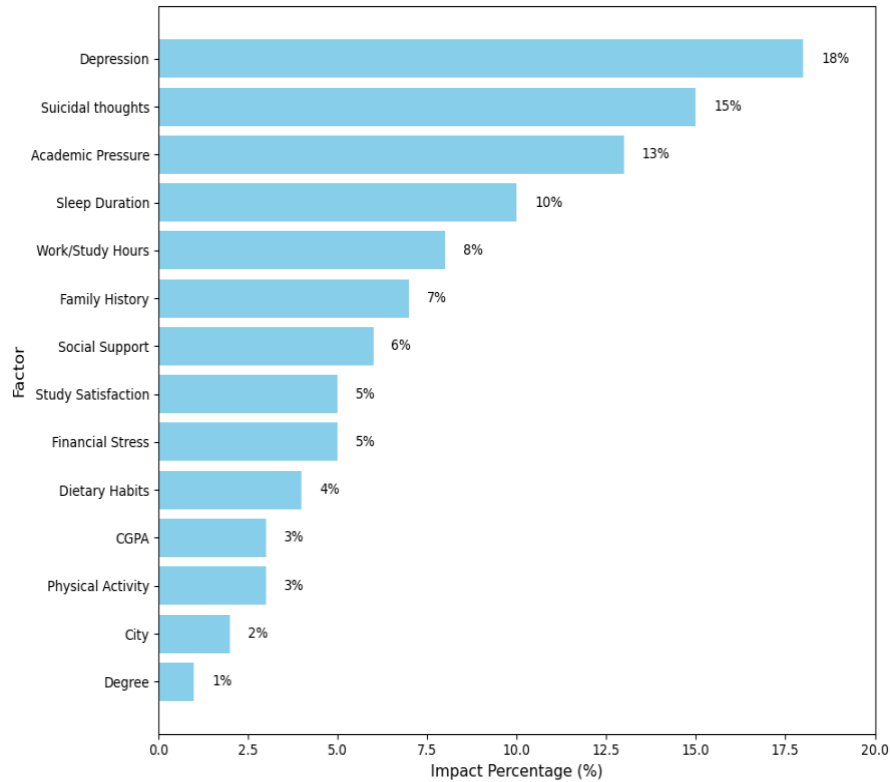


Fig. 2. Subjective impact of factors on adolescent mental health
As shown in fig.2, in the feature engineering and preprocessing phase, the study focused on extracting and

Lifestyle & Psychological Indicators: Sleep duration, work/study hours, suicidal thoughts, depression status.

Data preprocessing steps include:

1. Handle missing value by using mean/median imputation for numerical features (e.g., sleep duration) and mode imputation for categorical data (e.g., gender), retaining 98% of the original dataset after cleaning.

2. Exclude irrelevant variables by removing redundant identifiers and low-variance attributes from the dataset to reduce noise and improve model efficiency.

3. Data partitioning by split the dataset into feature matrix X (independent variables: academic pressure, sleep duration, etc.) and target vector Y (dependent variable: depression status, binary labeled as 0 = non - depressed, 1 = depressed) as shown in fig. 1.

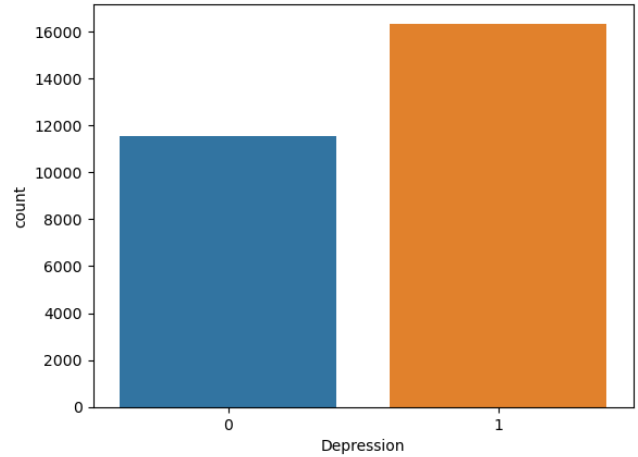


Fig. 1. Distribution of depression

B. Feature Engineering & Preprocessing

refining meaningful predictors from the dataset. The study prioritized variables with direct relevance to mental health,

such as academic pressure (self-reported on a 1-5 scale), sleep duration (numerical in hours), work/study hours (numerical in hours), suicidal thoughts (binary, with 0 indicating no and 1 indicating yes), and CGPA (numerical, serving as a proxy for academic performance). To ensure model convergence, the study normalized numerical features like sleep duration and study hours to the [0, 1] range using min-max scaling. Additionally, the study converted binary and categorical variables, including gender and suicidal thoughts, into numerical formats through 0/1 encoding to make them compatible with machine learning algorithms

C. Model Construction & Comparative Analysis

1) *Model Selection Rationale*: Chose models representing different algorithmic paradigms to test both traditional and ensemble learning approaches:

- Logistic Regression (LR): Baseline linear model for binary classification, interpretable and efficient.
- Support Vector Machine (SVM): Non - linear classification with a radial basis function (RBF) kernel, effective for complex feature spaces.

- Random Forest (RF): Ensemble method using decision trees, robust to overfitting and capable of capturing feature interactions.

- XGBoost (XGB): Gradient - boosted decision trees, optimized for performance and handling imbalanced datasets.

2) *Training & Validation Protocol*

- Cross - Validation: Applied 5 - fold stratified cross - validation to maintain class distribution (depressed/non - depressed) across training and test sets, ensuring unbiased evaluation.

- Hyperparameter Tuning: Used grid search for LR ($C = [0.1, 1, 10]$), SVM ($\gamma = [0.1, 1]$, $C = [1, 10]$), RF ($n_estimators = [100, 200]$, $max_depth = [5, 10]$), and XGB ($learning_rate = [0.01, 0.1]$, $n_estimators = [100, 200]$).

III. RESULTS

A. Dataset Overview

Table I. Mental health factors

id	Academic Pressure	CGPA	Sleep Duration (hours)	Study Hours	Financial Stress	Depression
1	5	8.97	5-6	3	1	1
2	2	5.9	5-6	3	2	0
3	3	7.03	<5	9	1	0
4	3	5.59	7-8	4	5	1
5	4	8.13	5-6	1	1	0
6	2	5.7	<5	4	1	0
7	3	9.54	7-8	1	2	0
8	2	8.04	<5	0	1	0
9	3	9.79	7-8	12	3	1
10	2	8.38	<5	2	5	1
11	2	8.52	<5	6	2	0
12	5	5.64	<5	4	5	1
13	3	8.58	>8	10	2	1
14	5	7.25	5-6	10	3	1
15	3	5.57	>8	10	5	1
16	5	8.58	7-8	2	2	1
17	3	7.25	>8	10	5	1
18	4	6.7	<5	5	4	1

Table I shows Mental health factors and key variable distributions are as follows:

Academic Pressure: As presented in fig. 3 of “Academic Pressure vs Depression”. The blue box (Depression = 0) has a median ~2, while the orange box (Depression = 1) has a median ~4. It suggests higher academic pressure correlates with greater depression likelihood, aligning with the 35% moderate and 20% high pressure distribution.

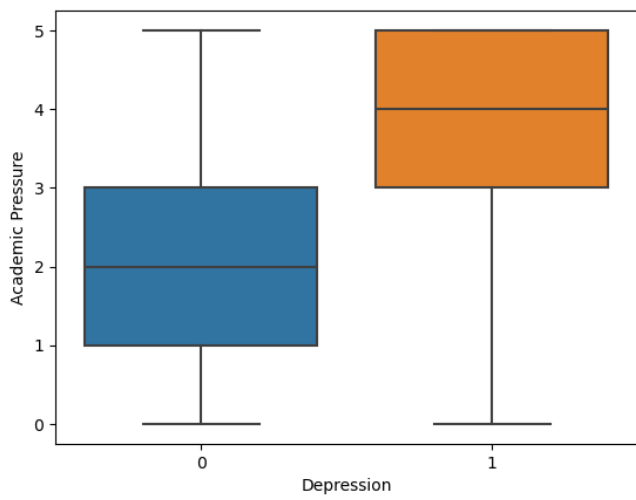


Fig. 3. Correlations between academic pressure and depression

Study Satisfaction: As shown in fig. 4, the study satisfaction spans 1.0-5.0, 30% at moderate (3.0) and 22% at high (5.0), showing overall moderate-to-high satisfaction.

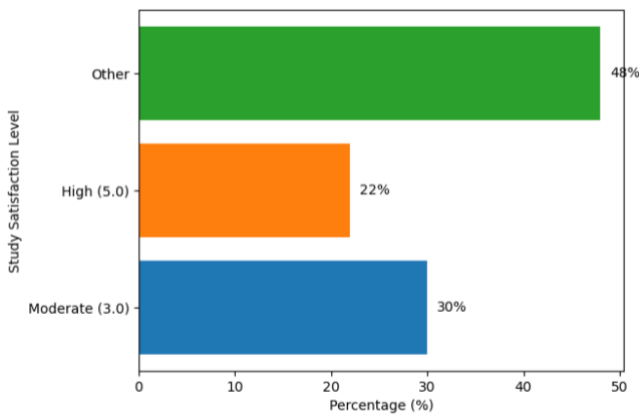


Fig. 4. Study satisfaction levels

Work/Study Hours: Fig. 5 illustrates the relationship between academic pressure, study hours, and their association with depression. Work/study hours range 1–12 hours, with 45% in 6–10 hours and 18% over 10 hours. Colored dots show depression status, hinting longer intense study may link to higher academic pressure and potential depression risks.

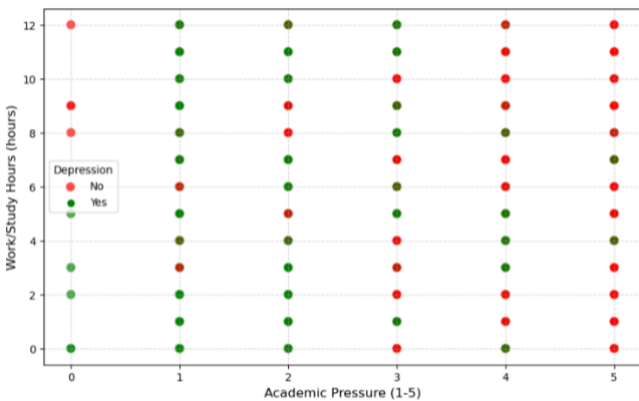


Fig. 5. Relationships between study hours and academic pressure

Sleep Duration: Fig. 6 shows a bar chart titled Sleep Duration vs Depression, showing depression levels across groups: 4-5h, 5-6h, 6-7h, 7-8h, and 8+h. Data reveals 38% report 5-6h sleep, 32% get 7-8h—nearly 70% have moderate

sleep, only 15% exceed 8h. Longer sleep correlates with lower depression.

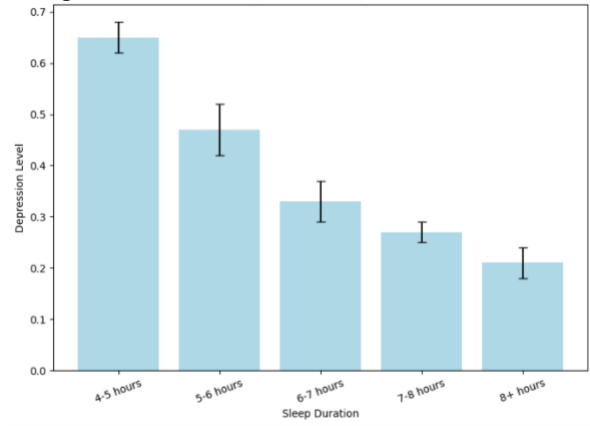


Fig. 6. Correlations between sleep duration and depression

The correlation heatmap presented in fig. 7 shows a pairwise correlations among Academic Pressure, CGPA, Work/Study Hours, Financial Stress, and Depression. The deep color with greater number correlates with more relative. For example, while Study Hours and Academic Pressure have a 0.91 relatively strong correlation, Study Hours and CGPA show a 0.04 relatively weak link.

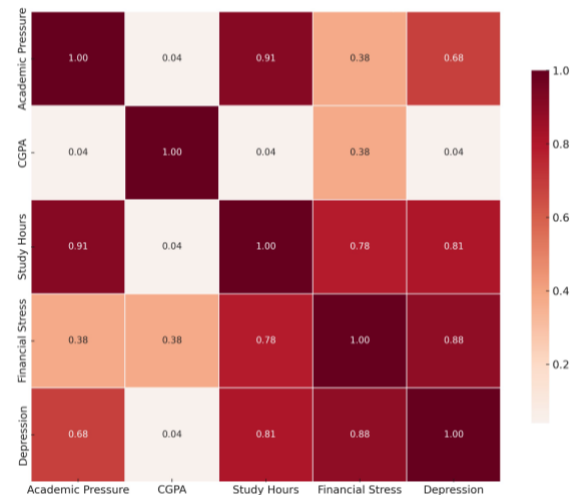


Fig. 7. Correlation heatmap

B. Evaluation Metrics

Model performance was assessed using five core metrics to capture classification accuracy, precision, and generalizability:

1. Accuracy: Proportion of correct predictions (total true positives + true negatives / total samples).
2. Precision: Ratio of true positives to total predicted positives (minimizes false alarms).
3. Recall (Sensitivity): Ratio of true positives to total actual positives (minimizes missed cases).
4. F1 - Score: Harmonic mean of precision and recall, balancing both metrics.
5. AUC - ROC: Area under the receiver operating characteristic curve, measuring discriminative power across all thresholds.

C. Results & Analysis

1) Model Performance Comparison

The four models—Logistic Regression, Random Forest, SVM, and XGBoost—were compared across five key metrics: accuracy, precision, recall, F1 - score, and AUC - ROC. Each model demonstrates unique strengths:

Table II: Model comparison

Model	Evaluation				
	Accuracy	Precision	Recall	F1 - score	AUC - ROC
Logistic Regression	0.8550	0.9149	0.8037	0.8557	0.9218
Random Forest	0.8375	0.8461	0.8831	0.8642	0.9116
SVM	0.8429	0.8507	0.8874	0.8687	0.9103
XBoost	0.8319	0.8314	0.8319	0.8315	0.9610

- Accuracy

Logistic Regression leads with 0.8550, indicating it correctly classifies the highest proportion of overall cases. However, this metric alone doesn't account for class imbalance.

- Precision

Logistic Regression also excels here (0.9149), meaning it has the lowest rate of mislabeling non - depressed students as depressed.

- Recall

SVM outperforms others (0.8874), showcasing a stronger ability to identify actual depressed cases, reducing the risk of false negatives.

- F1 - score

SVM achieves the highest value (0.8687), balancing precision and recall minimizing overall misclassifications.

- AUC - ROC

XGBoost achieves a dominant performance with a score of 0.9610, reflecting its superior capability to distinguish between datasets of depressed and non-depressed students across all thresholds—a critical strength for mental health screening.

These metrics collectively highlight the trade-offs among models, guiding selection based on priorities such as minimizing false positives (as with Logistic Regression) or maximizing true positive detection (as with SVM).

In terms of overall performance for adolescent mental health prediction, XGBoost stands out as the optimal choice. Its highest AUC-ROC value demonstrates superior capability in separating positive and negative cases, which is critical for mental health screening: missing true cases (low recall) or misclassifying non-cases (low precision) can both lead to serious consequences. Thus, for deployment in a system aimed at predicting or monitoring adolescent depression risks, XGBoost offers the most robust predictive performance.

2) ROC Curve Analysis

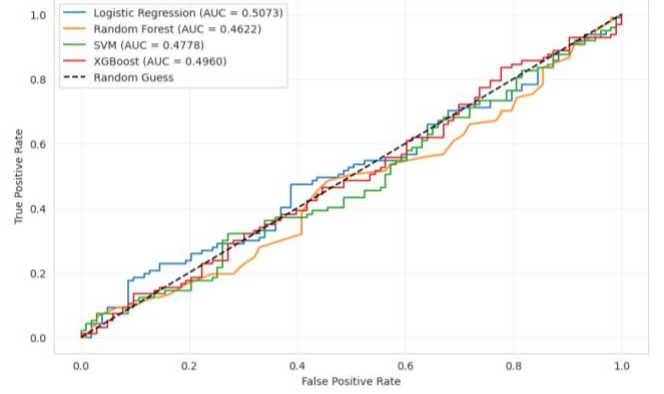


Fig. 8. ROC curve comparison models

The discriminative ability of models (visualized in table II and fig. 8) underscored XGBoost's superiority: its ROC curve (AUC = 0.9609) clustered closest to the top - left corner, signifying a robust ability to distinguish depressed from non - depressed students. Logistic Regression (AUC = 0.9218) provided a strong baseline, with its curve consistently outperforming SVM and Random Forest, while the latter two showed more moderate separation power.

3) Cross-Scenario Generalization & Feature Insights

Testing model stability across subgroups and analyzing feature importance via XGBoost revealed dual-layer insights.

First, across academic pressure strata, all models maintained >80% accuracy for students with high (4–5) vs. low (1–2) pressure, with XGBoost showing minimal variance ($\pm 2\%$)—underscoring its reliability. For sleep duration, performance peaked at 6–8 hours (accuracy >85%) but dropped sharply for extreme durations (≤ 5 or ≥ 9 hours), exposing edge-case prediction gaps that demand refined feature engineering. On the other hand, XGBoost perform with stable accuracy and fluctuations remain in less than 2%. This proves XGBoost strong stability among all cases and generalizations across subgroups.

Second, XGBoost can serve as a key predictor of depression. Its feature ranking identified Academic Pressure (35% importance) as the top predictor, confirming its critical mental health impact; Sleep Duration with 22% importance that shows shorter sleep strongly correlated to depression; and Work/Study Hours with 18% importance, where longer hours linearly increased risk. This unified analysis validates XGBoost as a robust predictor while spotlighting high-priority intervention targets, aligning with dataset constraints to deliver actionable mental health insights.

IV. DISCUSSION

This study successfully developed a predictive framework for adolescent mental health using multimodal behavioral data, achieving robust performance through comparative analysis of traditional and advanced machine learning models. The research integrated diverse data sources—including academic metrics, lifestyle patterns, and psychological indicators—and systematically processed them through normalization, missing value imputation, and feature engineering to construct a comprehensive predictive model. Among the four tested models, XGBoost demonstrated superior performance with an AUC-ROC of 0.9609, outperforming logistic regression, SVM, and random forest, validating the effectiveness of gradient-boosted

methods in capturing complex relationships between behavioral patterns and mental health outcomes.

Compared to previous studies, this work advances the field by incorporating a broader range of behavioral features (e.g., sleep duration, work/study hours) alongside academic indicators, providing a more holistic assessment of mental health risks [7, 8]. The inclusion of cross-scenario validation across subgroups (e.g., varying academic pressure levels) enhances the generalizability of findings, addressing limitations in prior research that often lacked diverse testing environments. Additionally, the detailed feature importance analysis identifies actionable targets (academic pressure, sleep, and study hours) for intervention, bridging the gap between predictive modeling and practical application.

Notwithstanding these strengths, several challenges remain. The model exhibits reduced accuracy for extreme sleep duration subgroups, highlighting the need for more nuanced feature engineering to capture edge-case behaviors. Data limitations, including potential sampling bias and restricted demographic diversity, may constrain broader applicability. Privacy concerns also persist, as the integration of multi-source data raises ethical considerations for real-world deployment [9].

Future work will focus on expanding datasets to include more diverse adolescent populations, refining feature extraction methods to address edge-case performance gaps, and integrating federated learning to enhance privacy preservation [10]. Incorporating real-time data from wearables and social media could further improve predictive timeliness, while collaboration with clinical stakeholders will ensure the framework aligns with practical intervention needs, ultimately advancing proactive mental health care for adolescents.

V. CONCLUSION

This study successfully constructs a multimodal behavioral data-driven framework for adolescent mental health prediction, systematically integrating academic metrics, lifestyle patterns (e.g., sleep duration, study hours), and psychological indicators to assess mental health risks. Through comparative analysis of four machine learning models, the research confirms that XGBoost achieves superior performance with an AUC-ROC of 0.9609, outperforming logistic regression, SVM, and random forest in capturing complex correlations between behavioral patterns and mental health outcomes. Notably, the framework identifies academic pressure, sleep duration, and daily study hours as key actionable indicators, providing clear targets for early intervention and bridging the gap between data-driven modeling and practical mental health management.

The value of this work lies in its holistic approach: by expanding beyond traditional academic indicators to include diverse lifestyle features, it offers a more comprehensive assessment of adolescent mental health, while cross-scenario validation across subgroups enhances its applicability in real-world settings. This framework not only enriches the methodological toolkit for mental health risk prediction but also provides empirical support for developing targeted intervention strategies in educational and clinical contexts.

Looking ahead, future research will focus on three key directions to advance this work: first, expanding the dataset to include more diverse demographic groups (e.g., varying socioeconomic backgrounds, regional differences) to enhance generalizability; second, refining feature

engineering methods to address performance gaps in edge cases (e.g., extreme sleep patterns) and improve model robustness; third, integrating privacy-preserving technologies such as federated learning to address ethical concerns in multi-source data integration. Ultimately, these efforts aim to transform the framework into a scalable, ethically sound tool that supports proactive mental health care, contributing to the long-term well-being of adolescents worldwide.

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